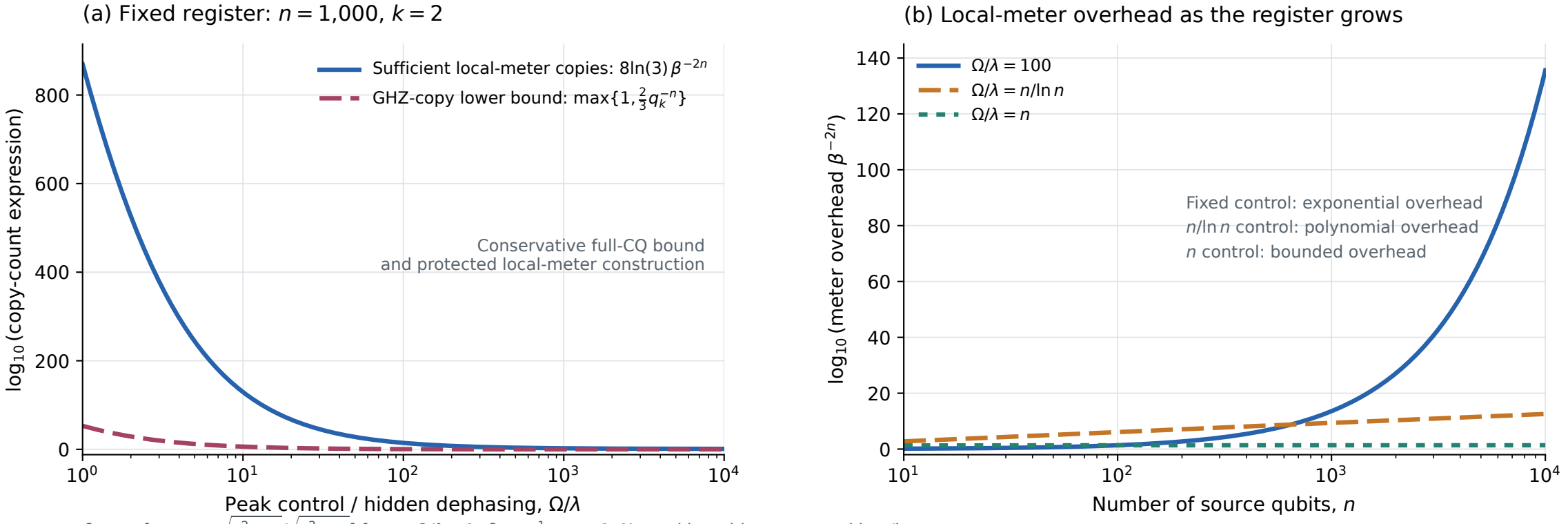
Resource bounds for acquiring global coherence

Initially unencoded sources with independent hidden dephasing

Bounds from distinct constructions; the plotted copy counts are not asserted optimal.



$\beta = \exp[-\arctan\sqrt{r^2-1}/\sqrt{r^2-1}\,]$ for $r = \Omega/\lambda > 1$; $\beta = e^{-1}$ at $r = 1$. Natural logarithms are used in $n/\ln n$.

Panel (b) excludes the constant sufficient-copy prefactor. Counts are unrounded; meter access and preparation costs are outside these curves.